다른 페이지로

react-router-dom

권기웅

1. 일단 준비

1.1 라이브러리 추가

프로젝트 만들고 나서, yarn add react-router-dom

```
Windows PowerShell
D:₩Kwon₩wsReact₩r11_page>yarn add react-router-dom
yarn add v1.22.19
[1/4] Resolving packages...
[2/4] Fetching packages...
[3/4] Linking dependencies...
warning " > @testing-library/user-event@13.5.0" has unmet peer dependency "@testing-library/dom
@>=7.21.4".
warning "react-scripts > eslint-config-react-app > eslint-plugin-flowtype@8.0.3" has unmet peer
 dependency "@babel/plugin-syntax-flow@^7.14.5".
warning "react-scripts > eslint-config-react-app > eslint-plugin-flowtype@8.0.3" has unmet peer
 dependency "@babel/plugin-transform-react-jsx@^7.14.9".
warning "react-scripts > react-dev-utils > fork-ts-checker-webpack-plugin@6.5.2" has unmet peer
 dependency "typescript@>= 2.7".
warning "react-scripts > eslint-config-react-app > @typescript-eslint/eslint-plugin > tsutils@3
.21.0" has unmet peer dependency "typescript@>=2.8.0 || >= 3.2.0-dev || >= 3.3.0-dev || >= 3.4.
0-dev || >= 3.5.0-dev || >= 3.6.0-dev || >= 3.6.0-beta || >= 3.7.0-dev || >= 3.7.0-beta".
[4/4] Building fresh packages...

success Saved lockfile.
success Saved 2 new dependencies.
info Direct dependencies
└─react-router-dom@6.4.2
info All dependencies
├─react-router-dom@6.4.2
└─react-router@6.4.2
Done in 5.03s.

D:₩Kwon₩wsReact₩r11_page>yarn start
```

1.2 적용

index.js

```python
import React from "react";
import ReactDOM from "react-dom/client";
import "./index.css";
import App from "./App";
import reportWebVitals from "./reportWebVitals";
import { BrowserRouter } from "react-router-dom";

const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(
  <BrowserRouter>
    <App />
  </BrowserRouter>
);

// If you want to start measuring performance in your app, pass a function
// to log results (for example: reportWebVitals(console.log))
// or send to an analytics endpoint. Learn more: https://bit.ly/CRA-vitals
reportWebVitals();
```

2. 다른 페이지로

2.1 기본 이동

2.1.1 페이지들

first.js

```
import React from "react";

const First = () => {
  return <h1>first</h1>;
};

export default First;
```

second.js

```
import React from "react";

const Second = () => {
  return <h2>second</h2>;
};

export default Second;
```

2.1.2 App.js

App.js

```
import { Route, Routes } from "react-router-dom";
import "./App.css";
import First from "./first";
import Second from "./second";

function App() {
  return (
    <Routes>
      <Route path="/" element={<First />} />
      <Route path="/se.cond" element={<Second />} />
    </Routes>
  );
}

export default App;
```

/, /se.cond 로 요청해서 확인

2.2 <a>

2.2.1 일단

<a>로 해도 되고...

2.2.2 <a> 대신

first.js

```
import React from "react";
import { Link } from "react-router-dom";

const First = () => {
  return (
    <>
      <h1>first</h1>
      <Link to="/se.cond">second로</Link>
    </>
  );
};

export default First;
```

2.3 요청주소 - useParams()

2.3.1 페이지들

second.js

```
import React from "react";
import { Link } from "react-router-dom";

const Second = () => {
  return (
    <>
      <h2>second</h2>
      <Link to="/thir.d/감자/500">감자, 500</Link>
      <br />
      <Link to="/thir.d/고구마/600">고구마, 600</Link>
    </>
  );
};

export default Second;
```

third.js

```
import React from "react";
import { useParams } from "react-router-dom";

const Third = () => {
  const snackInfo = useParams();

  return (
    <>
      <h1>{snackInfo.name}</h1>
      <h2>{snackInfo.price}</h2>
    </>
  );
};

export default Third;
```

2.3.2 App.js

App.js

```
import { Route, Routes } from "react-router-dom";
import "./App.css";
import First from "./first";
import Second from "./second";
import Third from "./third";

function App() {
  return (
    <Routes>
      <Route path="/" element={<First />} />
      <Route path="/se.cond" element={<Second />} />
      <Route path="/thir.d/:name/:price" element={<Third />} />
    </Routes>
  );
}

export default App;
```

2.4 요청파라메터 - useSearchParams()

2.4.1 페이지들

third.js

```
import React from "react";
import { Link, useParams } from "react-router-dom";

const Third = () => {
  const snackInfo = useParams();

  return (
    <>
      <h1>{snackInfo.name}</h1>
      <h2>{snackInfo.price}</h2>
      <Link to="/four.th?txt=ㅋ&count=3">ㅋ, 3</Link>
      <br />
      <Link to="/four.th?txt=ㅎ&count=5">ㅎ, 5</Link>
    </>
  );
};

export default Third;
```

fourth.js

```
import React from "react";
import { useSearchParams } from "react-router-dom";

const Fourth = () => {
  const [p, setP] = useSearchParams();
  const t = p.get("txt");
  const c = p.get("count");
  let cmt = "";
  for (let index = 0; index < c; index++) {
    cmt += t;
  }
  return <h1>{cmt}</h1>;
};

export default Fourth;
```

2.4.2 App.js

```
import { Route, Routes } from "react-router-dom";
import "./App.css";
import First from "./first";
import Fourth from "./fourth";
import Second from "./second";
import Third from "./third";

function App() {
  return (
    <Routes>
      <Route path="/" element={<First />} />
      <Route path="/se.cond" element={<Second />} />
      <Route path="/thir.d/:name/:price" element={<Third />} />
      <Route path="/four.th" element={<Fourth />} />
    </Routes>
  );
}

export default App;
```

2.5 클릭이벤트 말고 - useNavigate()

fourth.js

```
import React from "react";
import { useNavigate, useSearchParams } from "react-router-dom";

const Fourth = () => {
  const [p, setP] = useSearchParams();
  const t = p.get("txt");
  const c = p.get("count");
  let cmt = "";
  for (let index = 0; index < c; index++) {
    cmt += t;
  }

  const navi = useNavigate();
  return (
    <>
      <h1>{cmt}</h1>
      <h2
        onDoubleClick={() => {
          navi(-2);
        }}
      >
        두 번 뒤로
      </h2>
      <h2
        onDoubleClick={() => {
          navi("se.cond");
        }}
      >
        second로
      </h2>
    </>
  );
};

export default Fourth;
```

2.6 redirect - Navigate Component

hi.js

```
import React from "react";

const Hi = () => {
  return <h1>ㅎㅇ</h1>;
};

export default Hi;
```

first.js

```
import React from "react";
import { Link, Navigate } from "react-router-dom";

const First = () => {
  const r = Math.random();
  if (r > 0.5) {
    return <Navigate to="/h.i" replace={true} />;
  }
  return (
    <>
      <h1>first</h1>
      <Link to="/se.cond">second로</Link>
    </>
  );
};

export default First;
```

App.js

```
import { Route, Routes } from "react-router-dom";
import "./App.css";
import First from "./first";
import Fourth from "./fourth";
import Hi from "./hi";
import Second from "./second";
import Third from "./third";

function App() {
  return (
    <Routes>
      <Route path="/" element={<First />} />
      <Route path="/se.cond" element={<Second />} />
      <Route path="/thir.d/:name/:price" element={<Third />} />
      <Route path="/four.th" element={<Fourth />} />
      <Route path="/h.i" element={<Hi />} />
    </Routes>
  );
}

export default App;
```

2.7 기타

notFound.js

```
import React from "react";

const NotFound = () => {
  return <h1>떼</h1>;
};

export default NotFound;
```

App.js

```
import { Route, Routes } from "react-router-dom";
import "./App.css";
import First from "./first";
import Fourth from "./fourth";
import Hi from "./hi";
import NotFound from "./notFound";
import Second from "./second";
import Third from "./third";

function App() {
  return (
    <Routes>
      <Route path="*" element={<NotFound />} />
      <Route index element={<First />} />
      <Route path="/se.cond" element={<Second />} />
      <Route path="/thir.d/:name/:price" element={<Third />} />
      <Route path="/four.th" element={<Fourth />} />
      <Route path="/h.i" element={<Hi />} />
    </Routes>
  );
}

export default App;
```

3. 프레임

3.1 페이지들

home.js

```
import React from "react";

const Home = () => {
  return <h1>1회차</h1>;
};

export default Home;
```

students.js

```
import React from "react";

const Students = () => {
  let studentList = [];
  for (let index = 1; index < 22; index++) {
    studentList = studentList.concat(<h3 key={index}>이상한 사람 {index}</h3>);
  }
  return <>{studentList}</>;
};

export default Students;
```

banjang.js

```
import React from "react";

const Banjang = () => {
  return <h1>가장 이상한 사람</h1>;
};

export default Banjang;
```

3.2 layout

layout.js

```
import React from "react";
import { Link, Outlet } from "react-router-dom";

const Layout = () => {
  const css = {
    width: "1200px",
    marginLeft: "auto",
    marginRight: "auto",
  };
  return (
    <table border={1} style={css}>
      <tr>
        <td align="center">
          <h1>
            블록체인 네트워크 보안을 적용한 AI 서비스 기반 풀스택 개발자 과정
          </h1>
        </td>
      </tr>
      <tr>
        <td align="center">
          <Link to="/">홈</Link>     
          <Link to="/stu.dents">학생들</Link>     
          <Link to="/ban.jang">반장</Link>
        </td>
      </tr>
      <tr>
        <td align="center">
          <Outlet />
        </td>
      </tr>
    </table>
  );
};

export default Layout;
```

3.3 App.js

```
import { Route, Routes } from "react-router-dom";
import "./App.css";
import Banjang from "./banjang";
import Home from "./home";
import Layout from "./layout";
import Students from "./students";

function App() {
  return (
    <Routes>
      <Route element={<Layout />}>
        <Route path="/" element={<Home />} />
        <Route path="/stu.dents" element={<Students />} />
        <Route path="/ban.jang" element={<Banjang />} />
      </Route>
    </Routes>
  );
}

export default App;
```

3.4 활성 메뉴 디자인 - NavLink Component

layout.js

```
import React from "react";
import { NavLink, Outlet } from "react-router-dom";

const Layout = () => {
  const css = {
    width: "1200px", marginLeft: "auto", marginRight: "auto"
  };
  const css2 = {
    color: "red",
    fontSize: "20pt",
  };
  return (
    <table border={1} style={css}>
      <tr>
        <td align="center">
          <h1>
            블록체인 네트워크 보안을 적용한 AI 서비스 기반 풀스택 개발자 과정
          </h1>
        </td>
      </tr>
      <tr>
        <td align="center">
          <NavLink to="/index.go" style={({ isActive }) => (isActive ? css2 : undefined)}>홈</NavLink>     
          <NavLink to="/stu.dents" style={({ isActive }) => (isActive ? css2 : undefined)}>학생들</NavLink>     
          <NavLink to="/ban.jang" style={({ isActive }) => (isActive ? css2 : undefined)}>반장</NavLink>
        </td>
      </tr>
      <tr>
        <td align="center">
          <Outlet />
        </td>
      </tr>
    </table>
  );
};

export default Layout;
```

App.js

```
import { Route, Routes } from "react-router-dom";
import "./App.css";
import Banjang from "./banjang";
import Home from "./home";
import Layout from "./layout";
import Students from "./students";

function App() {
  return (
    <Routes>
      <Route element={<Layout />}>
        <Route index element={<Home />} />
        <Route path="/index.go" element={<Home />} />
        <Route path="/stu.dents" element={<Students />} />
        <Route path="/ban.jang" element={<Banjang />} />
      </Route>
    </Routes>
  );
}

export default App;
```